Practical No. 02

Simulation of ARP using Cisco Packet Tracer

Easiest revision notes, written for someone starting with zero networking knowledge

What This Practical Is About

In this practical you connect three computers to one hub, then you make one computer ping another one. Before the ping message can actually travel, the sending computer needs to learn the physical address of the receiving computer first. That learning process is called ARP. Packet Tracer lets you watch this happen step by step in slow motion using Simulation Mode. Everything below explains each term and step slowly so nothing feels confusing.

Key Terms Explained First

IP address. A unique number given to a device on a network, used to identify it and send data to it, similar to a house address.

MAC address. A unique physical address built into every network card. While an IP address can be seen as your house address, a MAC address is more like your actual fingerprint, it belongs permanently to that one piece of hardware.

ARP, Address Resolution Protocol. A protocol whose only job is to find out the MAC address of a device when you already know its IP address. Computers know each other by IP address, but data on a local network actually travels using MAC addresses, so ARP is the translator in between.

ARP table, also called ARP cache. A small list stored in a computer's memory that keeps a record of which IP address belongs to which MAC address. Once a computer learns an answer through ARP, it saves it here so it does not have to ask again every time.

ICMP, Internet Control Message Protocol. The protocol used for the ping command. It is used to check whether another device can be reached over the network and how the connection is behaving.

Ping. A simple test message sent from one device to another to check if it is reachable. If the other device is alive and reachable, it sends a reply back.

Hub. An old and simple networking device. When it receives data from one connected device, it copies that data out to every other connected device, without checking who the data was actually meant for. It has no intelligence at all.

Switch. A smarter device compared to a hub. It learns the MAC address of every device connected to it and sends incoming data only to the specific device it is meant for, instead of copying it everywhere.

Broadcast. Sending a message to every device on the network at once, instead of to just one specific device. An ARP Request is always a broadcast, because the sender does not yet know who exactly holds the IP address it is asking about.

Unicast. Sending a message to exactly one specific device, not to everyone. An ARP Reply is a unicast, because by that point the replying device knows exactly who asked and sends the answer straight back to them only.

Packet and Frame. Small chunks of data that travel across a network, each one wrapped with some extra header information describing where it came from and where it is going. People often use the word packet loosely, but technically a frame is the version of this data used at the lower, hardware based level of networking.

OSI layers. A model that breaks down how a network works into separate stages, called layers, stacked from the most physical at the bottom to the most abstract at the top. This practical mainly touches three of them, Physical Layer, Data Link Layer and Network Layer, explained further below.

Simulation Mode. A special mode inside Packet Tracer that slows everything down and lets you click through each stage of a message travelling across the network, instead of it happening instantly like in Realtime Mode.

Traffic Generator. A tool inside Packet Tracer used to manually build and send a custom test message, such as a ping, from one chosen device to another, so you can control exactly what gets sent and watch it happen.

The Setup You Built

One hub connects three PCs together. Each PC is given its own IP address as shown below.

Device	IP Address

PC1	192.168.1.1
PC2	192.168.1.2
PC3	192.168.1.3

Using the Traffic Generator, a ping is sent from PC3 to PC1, meaning Destination is 192.168.1.1, Source is 192.168.1.3, Sequence number 1.

Two useful commands used during the practical are, **arp -a** which shows the current ARP table of a device, and **arp -d** which clears that ARP table so you can watch the whole learning process happen again from scratch.

The One Trick That Explains the Whole Practical

Only two devices ever actually talk to each other, PC3 which is sending the ping, and PC1 which is receiving it. PC2 does nothing the entire time, it simply notices a message is not meant for it and quietly drops it, that is exactly why PC2's ARP table stays empty at the end.

Everything that happens can be grouped into two rounds. Round one is ARP, where PC3 finds out PC1's MAC address. Round two is ICMP, which is the actual ping and its reply.

ASK, TELL, PING, REPLY

ASK, PC3 asks who owns this IP address, this is the ARP Request.
TELL, PC1 tells PC3 its MAC address, this is the ARP Reply.
PING, PC3 finally sends the real ping, this is the ICMP Request.
REPLY, PC1 answers the ping, this is the ICMP Reply.

Step Map, So You Never Lose Track

Match the click number you see in Packet Tracer to the row below to instantly know what is happening.

Click #	Direction	What Is Moving	In Simple Words
1	At PC3	ARP Request is created	PC3 thinks, I do not know PC1's MAC address, let me ask.
2	PC3 to Hub	ARP Request sent	The request leaves PC3 and reaches the hub.
3	Hub to all	Hub broadcasts	The hub copies the message out to every connected port, PC1 and PC2 both receive it.
4	At PC2	PC2 drops it	PC2 checks the IP address, sees it is not addressed to it, and simply ignores it, no ARP entry is created here.
5	At PC1	PC1 accepts it	PC1 checks the IP address, sees it matches its own, and saves PC3's information.
6	PC1 to Hub	ARP Reply sent	PC1 replies with its own MAC address.
7	Hub to PC3	Hub forwards the reply	The reply travels back and reaches PC3.
8	At PC3	PC3 saves the MAC address	PC3's ARP table now has an entry for PC1.
9	PC3 to PC1	ICMP Ping sent	Now that PC3 knows the MAC address, the actual ping finally goes out.
10	PC1 to PC3	ICMP Ping Reply	PC1 answers the ping.
11	Check all three	Compare ARP tables	PC1 and PC3 both now have an entry for each other, PC2's table stays empty.

In short, clicks one through eight belong to the ARP round, which is about finding the MAC address. Clicks nine and ten belong to the ICMP round, which is the actual ping. Click eleven is just the final check.

Network Layers of Each Device and Protocol

This is a very common viva question, so it deserves its own clear table.

Device / Protocol	OSI Layer
Hub	Physical Layer, Layer 1
ARP	Data Link Layer, Layer 2
Switch	Data Link Layer, Layer 2
ICMP	Network Layer, Layer 3

Memory trick, the layer number goes up as the device gets smarter. A hub is the dumbest, it just repeats electrical signals, so it sits at Layer 1. A switch and ARP both understand and work with MAC addresses, so they sit at Layer 2. ICMP understands IP addresses, which is a more advanced concept, so it sits at Layer 3.

> A normal switch is Layer 2. A special advanced type called a Layer 3 switch can also work at Layer 3, but for this practical, just remember plain Switch equals Layer 2.

Viva Questions, Including Trickier Ones

Q1. What is ARP?
ARP, Address Resolution Protocol, is used to find the MAC address of a device when only its IP address is known.

Q2. Why is ARP needed at all?
Because devices on a local network actually deliver data using MAC addresses, not IP addresses, so ARP finds the correct MAC address before real data can be sent.

Q3. Why did an ARP message appear before the ping?
Because PC3's ARP table was empty at the start, it did not yet know PC1's MAC address.

Q4. Why does PC2 never get an ARP entry?
PC2 was never actually part of the conversation between PC3 and PC1, so it never had a reason to learn any MAC address.

Q5. Which comes first, ARP or ICMP?
ARP always comes first, ICMP, the actual ping, follows only once the MAC address is known.

Q6. What does a hub do with a broadcast frame?
It floods the frame out to every connected port except the one it arrived on, since a hub has no way of checking addresses.

Q7. What is the difference between an ARP Request and an ARP Reply?
An ARP Request is a broadcast sent to everyone, asking who owns a certain IP address. An ARP Reply is a unicast, sent directly and only to the device that asked.

Q8. Which command shows the ARP table?
arp -a

Q9. Which command clears the ARP table?
arp -d

Q10. Does ARP belong to Layer 2 or Layer 3?
ARP sits between the two, it maps a Layer 3 address, the IP address, to a Layer 2 address, the MAC address, but it is usually classified as a Data Link Layer, Layer 2, protocol.

Q11. What happens if two devices already know each other's MAC address?
No new ARP request is sent, the device simply uses the existing entry already saved in its ARP table.

Q12. How is a hub different from a switch in this context?
A hub always broadcasts to every port, while a switch learns MAC addresses over time and sends data only to the correct, specific port.

Q13. Which OSI layer does ARP work at?
Data Link Layer, Layer 2.

Q14. Which OSI layer does ICMP work at?

Network Layer, Layer 3.

Q15. Which OSI layer does a hub work at?

Physical Layer, Layer 1.

Q16. Which OSI layer does a switch work at?

Data Link Layer, Layer 2.

Important Things to Remember for Future Labs and Exams

- ARP only works inside the same local network. It cannot be used to find the MAC address of a device on a completely different network across the internet.
- ARP table entries are temporary, they expire automatically after some time, known as the ARP cache timeout, and get refreshed by asking again when needed.
- ARP Request equals broadcast, sent to everyone. ARP Reply equals unicast, sent to one device only. This exact point is asked very often.
- A hub floods data to every port, a switch is smarter and sends data only to the required port, this comparison is frequently asked alongside ARP questions.
- Gratuitous ARP is when a device announces its own IP to MAC mapping to everyone without being asked, mainly used to update other devices or to detect if another device is wrongly using the same IP address.
- RARP, Reverse ARP, is the opposite idea, it finds an IP address when only the MAC address is known. It is an older protocol, sometimes asked as a trick question.
- In real networks all of this happens in a few milliseconds automatically, Packet Tracer's Simulation Mode simply slows it down artificially so students can study each step clearly.

Quick recall order: Setup, then the trick with its two rounds, then the step map, then the viva questions.